

SHARAN KALAMOHAN

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PROFESSIONAL SUMMARY

MSc Automotive & Motorsport Engineering student (on track for Distinction) with a BEng in Mechanical Engineering (2:1). Experienced through a 12-month placement at Renishaw plc, where I developed a gauging system awarded Renishaw's Innovation Product of the Year 2024. Experienced across combustion and electrified powertrains, with a consistent record of validating simulation against measured data (FEA to within 2.2–7.1%, lap-time simulation to 0.7%). Engineering Technical Director of a Formula Student Concept Class team developing an electric concept car, owning the master CAD and vehicle-level targets across four subsystem groups.

KEY SKILLS

- **CAD & PLM:** Siemens NX; SolidWorks; Autodesk Inventor; Teamcenter PLM; 3D scanning and reverse engineering.
- **Simulation:** MATLAB & Simulink; Ricardo WAVE; Ansys Fluent CFD; Abaqus CAE; IPG CarMaker.
- **Manufacturing:** GibbsCAM; CNC milling and turning; Metal AM; CMM programming (Renishaw MODUS); VCDS data logging; DFM/DFA.

EDUCATION

MSc Automotive & Motorsport Engineering — Brunel University London

Sep 2025 – Sep 2026

On track for Distinction — Vehicle Dynamics & CAD (87%); Vehicle Propulsion Systems & Technologies (71%); Racing Vehicle Design (70%).

- **Dissertation:** Hydrogen-engine air intake study via coupled 1D–3D WAVE and Ansys modelling to quantify volumetric efficiency gains.
- **Coursework:** Hybrid powertrain architecture study: modelled P0/P1/P2 configurations in Simulink Powertrain Blockset (73%); MATLAB Formula 4 suspension optimisation at Silverstone (83%); Rally-stage telemetry analysis and lap time simulation with DC motor control (78%).

BEng (Hons) Mechanical Engineering — University of Nottingham

Sep 2021 – Jun 2025

2:1, including a 12-month industrial placement at Renishaw plc — Powertrain Engineering (91%); Automotive Technology (71%).

- **Dissertation (71%):** validated torsional FEA of concentric DCT shafts to within **2.2–7.1%** of classical theory and quantified material selection through a weighted performance index for lightweighting: 50B46 **tripled** inner-shaft safety factor (2.04 to 6.23) for a **+1.35%** cost increase.

PROFESSIONAL ENGINEERING EXPERIENCE

Mechanical Engineering Placement Student — Renishaw plc, Wotton-under-Edge

Aug 2023 – Aug 2024

Equator X gauging system — released to market and internally named **Renishaw's Innovation Product of the Year in 2024**.

- Identified a bonding-stage bottleneck capping strut output; designed and machined a multi-slot adhesive bonding jig that raised output **5×**.
- Cut bond-line thickness spread by **93%** through adhesive process iteration, achieving the CTE stability required for the calibration software.
- Planned and ran test programmes (adhesive life, thermal expansion, tensile); reports drove adhesive selection and production feasibility.
- Owned parts concept to verification: Siemens NX/Teamcenter CAD, GibbsCAM toolpaths, 3-axis machining, CMM inspection.
- Coordinated with Manufacturing and New Product Integration teams to transition the gauging system into full-scale production.

Engineering Technical Director — Brunel Masters Motorsport (Formula Student UK)

Oct 2025 – May 2026

IMechE's FSUK Concept Class — leading **30 engineers** across four groups developing an electric Formula Student concept car

- Held technically accountable for Powertrain, Chassis, Suspension and Driver Controls groups; owned master CAD and design specification.
- Set and arbitrated vehicle-level targets (**280 kg, 453 V nominal, 40:60 front/rear**), adopted by all four groups for integrated development.
- Chaired weekly design reviews resolving cross-group interface clashes; reported progress against deadlines to supervisors.
- Delivered the May Concept Class submission — full vehicle CAD and specifications, demonstrating rule compliance.

Motor Subsystem Lead — Brunel Masters Motorsport (Formula Student) / MSc Major Project

Sep 2025 – May 2026

- Developed a forward–backward quasi-static lap simulation in MATLAB for all four FS dynamic events, integrating a digitised 6,271-point Emrax efficiency map and a lumped-parameter thermal management model of the motor to quantify feasibility at the Silverstone event.
- Correlated predicted acceleration to within **0.7%** of the 2025 FSUK-winning 75 m time.
- Selected the Emrax 228 MV LC against powertrain and rule constraints; optimised final drive to **3.60** via a 144-run weighted sweep.
- Identified a **7 kW DC-bus rule breach** at 80 kW shaft power and implemented a torque cap ensuring rule compliance.
- Built an SoC-aware adaptive torque controller that recovered **94 s** of endurance time against a fixed derated map, preserving range.
- Demonstrated accumulator C-rate to be bottlenecking regenerative energy (**7.1%**), redirecting the powertrain team's next design priority.

PERSONAL PROJECTS

1D Engine Modelling & Vehicle Data Correlation — Self-directed

Jul 2026 – Present

- Logged full-throttle and part-load data (boost, mass air flow, intake air temperatures, lambda) from a turbocharged petrol engine via VCDS.
- Building a Ricardo WAVE air-path model on scaled turbocharger maps, correlating against logged data to evaluate vehicle performance.

ADDITIONAL EXPERIENCE & INTERESTS

Assistant Area Manager (part-time) — Mohan Retail Ltd, Amersham

Sep 2022 – Present

- Managed logistics, stock, staff and supplier communication across four family-run retail stores alongside full-time study.
- Introduced Excel stock rotation and dynamic pricing, cutting spoilage and increasing weekly revenue by **3.6%**.

Languages: English (Native), Tamil (Native) | **IMechE Associate Member** | **Interests:** music production; strength training; 5-a-side football